

Damiano Marsili

PhD student in Computing and Mathematical Sciences, Caltech, Pasadena, USA
Email, Personal Website, Google Scholar

EDUCATION

California Institute of Technology, Pasadena, USA
PhD Student, Computing and Mathematical Sciences

Sep 2023 — May 2028

Johns Hopkins University, Baltimore, USA
BS Computer Science
BA Mathematics

Aug 2020 — May 2023

PUBLICATIONS

Self-Verifying Vision Language Models (PDF)

In review, 2026

Damiano Marsili, Georgia Gkioxari

- Developed GenVer, a self-verification training framework that improves visual reasoning in VLMs without annotations.
- Showed that jointly training generation and verification improves reasoning, including +15.7% on 3D reasoning.

Same or Not? Enhancing Visual Perception in Vision-Language Models (PDF)

CVPR 2026 (**Highlight**)

Damiano Marsili, Aditya Mehta, Ryan Lin, Georgia Gkioxari

- Created TWIN, a large-scale dataset of 561K VQA queries designed to improve fine-grained understanding in VLMs.
- Introduced FGVQA, a benchmark suite of 12,000 queries that repurposes retrieval datasets for fine-grained VQA.
- Demonstrated that post-training on TWIN improves fine-grained understanding in VLMs, measured by an improvement of up to 19.3% on FGVQA, without compromising performance on general VQA.

No Labels, No Problem: Training Visual Reasoners with Multimodal Verifiers (PDF)

ICLR 2026

Damiano Marsili, Georgia Gkioxari

- Developed VALOR, an annotation-free post-training framework that uses multimodal verifiers to jointly improve visual reasoning via reinforcement learning and visual grounding through automated hard-negative mining.

Visual Agentic AI for Spatial Reasoning with a Dynamic API (PDF)

CVPR 2025

Damiano Marsili, Rohun Agrawal*, Yisong Yue, Georgia Gkioxari*

- Designed a training-free agentic visual programming approach, VADAR, that dynamically generates new skills in Python and significantly outperforms previous visual programming methods on spatial reasoning in 3D.
- Introduced Omni3D-Bench, a benchmark for 3D understanding with complex queries involving multiple reasoning steps.

WORK EXPERIENCE

Amazon Robotics

Applied Science Intern

Arlington, USA
May 2023 — Sep 2023

- Engineered a large multimodal spatial reasoning dataset composed of over 300,000 robotic grasp queries.
- Post-trained a Vision Language Model (VLM) to resolve spatial relationships for the task of targeted grasping.

Applied Physics Laboratory (JHU APL)

Research Assistant

Baltimore, USA
Aug 2022 — May 2023

- Worked on self-supervised training methods to train robots for gesture recognition using both synthetic and real data. Project funded by Army Research Labs (ARL).

Malone Center for Engineering in Healthcare, Johns Hopkins University

Research Intern

Baltimore, USA
Feb 2022 — May 2022

- Developed simulation environments used to train reinforcement learning agents for autonomous ventilators.

ACADEMIC SERVICE

Teaching Assistant

- Large Language & Vision Models (2026 Spring, Caltech).
- Object Oriented Software Engineering (2022 Spring, Johns Hopkins University)

Reviewer

- **Conferences:** CVPR (Outstanding Reviewer 2026), ICML (Gold Reviewer 2026), ECCV, NeurIPS, COLM
- **Workshops:** MUSI (CVPR 2026)

Tutoring

- Mathematics Tutor (Fall 2021, Learning Den - Johns Hopkins University).